

ADVANCED WATER SAFETY

Teacher Resource Guide

Year 9–10 HPE · Victorian Curriculum F–10 Version 2.0 · Companion to the Year 9–10 Student Workbook

- Year 9–10
- 5 Lesson Plans
- Complete Answer Key
- A–E Assessment Rubric
- Differentiation
- VC2HP10P02 · VC2HP10P08 · VC2HP10P09 · VC2HP10M01

Contents

- Curriculum Alignment & Achievement Standards
- Unit Overview & Learning Intentions
- Lesson Plans (Lessons 1–5)
- Student Workbook — Answer Key
- Assessment Rubric (A–E, 6 Criteria)
- Assessment Checklist (Summative Task)
- Differentiation & Inclusion
- Teacher Background Notes
- Resources & Sources

1 Curriculum Alignment & Achievement Standards

Code	Content Description	Strand	How this unit addresses it
VC2HP10P02	Refine, evaluate and adapt strategies for managing changes and transitions	Personal, Social & Community Health	Students investigate how peer influence, adolescent risk-taking behaviour, and social transitions affect exposure to drowning risk; explore social

			determinants using a structural lens
VC2HP10P08	Plan, rehearse and evaluate strategies (including first aid and CPR) for managing situations where their own or others' health, safety or wellbeing may be negatively impacted at home, school and in the community	Personal, Social & Community Health	Students plan and rehearse DRS ABCD emergency response; evaluate CPR and first aid options; analyse drowning prevention strategies using evidence and data
VC2HP10P08	Plan, rehearse and evaluate strategies (including first aid and CPR) for managing situations where their own or others' health, safety or wellbeing may be negatively impacted at home, school and in the community	Personal, Social & Community Health	Students design community interventions; audit school provision; evaluate accessibility of water safety programs for at-risk groups; propose targeted actions
VC2HP10P09	Critique health information, services and media messaging about relationships, lifestyle choices, health decisions and behaviours to evaluate their influence on their own, others' and community's attitudes and actions	Personal, Social & Community Health	Students evaluate credibility of prevention evidence from LSV, WHO, RLSSA and international sources; compare and apply findings to Victorian decisions
VC2HP10M01	Apply specialised movement concepts in aquatic contexts; analyse performance against criteria	Movement & Physical Activity	Students apply survival swimming theory and rescue technique analysis (pool sessions if available)

Achievement Standard Connections — Levels 9 & 10

By the end of Level 10, students are expected to:

- Critically analyse health information and the reliability of evidence in health contexts
- Plan and evaluate strategies that build individual and community capacity for health promotion
- Develop and evaluate complex health literacy skills including source evaluation
- Apply and refine specialised movement skills; analyse performance using data
- Propose evidence-based community health solutions with measurable outcomes

2 Unit Overview & Learning Intentions

Element	Detail
Unit Title	Advanced Water Safety: Data, Physiology & Prevention Policy

Year Level	Year 9–10 (Ages 14–16)
Duration	5 × 50-minute lessons (expandable with pool sessions via Pool Practical Companion)
Prerequisite	Year 7–8 Water Safety unit (or equivalent foundational knowledge of flags, DRS ABCD, SwimSafe rules)
Assessment	Workbook (Sections 1–5) + Summative Task (Section 6: policy brief, multimodal campaign, or school audit)
Key Departure from Y7–8	Emphasis shifts from knowledge and application to analysis, evaluation, and synthesis. Students are expected to use data, argue positions, and produce evidence-based products rather than identify and describe.

 Know / Understand / Do

KNOW

- Victorian drowning rates vs national/global
- Physiology of drowning and CPR
- Social determinants of drowning risk
- Prevention strategy evidence hierarchy
- International approaches (RLSSA, RLSS, Red Cross)

UNDERSTAND

- Why data alone doesn't prevent drowning
- How social factors amplify or reduce risk
- Why some prevention strategies work better than others
- The relationship between policy, community, and individual action

DO

- Calculate and interpret drowning rate data
- Evaluate prevention evidence critically
- Compare international approaches
- Design an evidence-based community product
- Argue a health policy position

 ENGAGE

8 MIN

- Open with: "Victoria had 54 drowning deaths in 2023–24. That's 0.78 per 100,000. Australia's national rate is 1.2 per 100,000. Should we be proud of this, or worried by it? Discuss." Pair discussion (2 min) → class share.
- Ask: "What social factors might explain why Victoria is BELOW the national rate on some metrics yet has particular problems with CALD communities and inland waterways?"
- Distribute workbooks — introduce the expectation shift: "At Year 9–10, describe is not enough. We analyse, evaluate, and justify."

 EXPLORE

15 MIN

- Students complete Workbook Activity 1.1 calculations independently: percentage calculation (Q1.1a), rate comparison (Q1.1b), and CALD over-representation analysis (Q1.1c).
- Circulate and prompt: "Show your working. What does the number MEAN, not just what it is?"
- If devices available: students access LSV Drowning Report to verify their figures and source their graph data for Q1.1d.

 EXPLAIN

12 MIN

- Teacher-led discussion: "CALD communities = 40% of Victorian drowning deaths, ~28% of population. What does over-representation mean? How do we explain it without stereotyping?"
- Introduce the concept of social determinants: access to swimming lessons, cultural norms around water, fishing risk, language barriers to safety messaging.
- Reference the LSV multicultural safety page (lsv.com.au/community/multicultural-safety/) as evidence that LSV is specifically trying to address this.

 ELABORATE

10 MIN

- Students complete data visualisation task (Q1.1d): bar graph of deaths by location type. Emphasise: labels, title, axes, interpretation — not just drawing bars.
- Pair check: swap workbooks and verify each other's calculations for Questions 1.1a–c.

 EVALUATE

5 MIN

- Exit ticket: "In one sentence: what is the SINGLE most important data insight from today's lesson, and why?"

 Teacher Note — CALD Discussion

Handle the CALD data discussion carefully. The goal is systemic analysis — why do structural factors (limited swim access, language barriers, fishing culture risk) create disproportionate

outcomes — not a commentary on any particular community. Frame as: "This data tells us the system hasn't served everyone equally, not that some groups are less careful."

2

Physiology of Drowning & CPR Mechanics

🕒 50 min

Section 2

VC2HP10PO8 · VC2HP10MO1

Classroom

🔥 ENGAGE

8 MIN

- Open with: "CPR generates only 25–30% of normal cardiac output. If normal output keeps you conscious and alive, why is 25–30% enough to bridge survival until an ambulance arrives?"
- Think-pair-share. Draw on whiteboard: normal cardiac cycle vs CPR cycle. Establish prior knowledge baseline.

🔍 EXPLORE

15 MIN

- Students complete Workbook Activity 2.1a (physiological sequence of drowning) and 2.1b (CPR physiology and 4-minute window).
- Key vocabulary to reinforce: hypoxia, laryngospasm, cardiac arrest, cerebral hypoxia, cardiac output, oxygenation.
- Circulate: ensure students are using the vocabulary correctly, not just repeating it.

🗨️ EXPLAIN

12 MIN

- Explain secondary drowning — a real but rare risk. Symptoms appear hours after a near-drowning event: coughing, laboured breathing, fatigue, chest pain. Always seek medical review after any submersion incident, even if the person appears fine.
- Walk through the 1-10-1 cold water survival rule with reference to Lake Eildon context. Ask: "Why does the same water temperature affect a 14-year-old vs a 45-year-old differently?" (Body composition, fitness level, clothing.)

✍️ ELABORATE

10 MIN

- Students complete Q2.1c (secondary drowning research) — requires device access.
- Class discussion: "What is the implication for parents, teachers, and coaches who supervise water activities? What should happen AFTER a near-miss?"

📊 EVALUATE

5 MIN

- Exit ticket: "Explain the 4-minute window to someone who has never heard of it — in plain language, not medical jargon."

 ENGAGE

8 MIN

- Open with the "Values Clarification" exercise: post four corners with signs — "Strongly Agree," "Agree," "Disagree," "Strongly Disagree." Read statements: "Males drown more because they're less careful than females." "CALD communities drown more because of language barriers."
- Students move to their corner and defend. Teacher challenges each position. Point: these are over-simplifications of a complex social issue. Introduce the concept of structural vs individual explanations.

 EXPLORE

15 MIN

- Students complete Workbook Activity 3.1a (social determinants table). Pairs — assign different determinants to each pair for jigsaw reporting later.
- Prompt: "What does the research actually say? What's the difference between correlation and causation?"

 EXPLAIN

12 MIN

- Jigsaw sharing: each pair reports on their assigned determinant (5 min). Teacher synthesises.
- Introduce the WHO Social Determinants of Health Framework: "Health outcomes are shaped by the conditions in which people are born, grow, live, work, and age."
- Connect to Victorian data: CALD over-representation, male risk-taking, rural/regional inland waterway risk.

 ELABORATE

12 MIN

- Students complete Q3.1b (3 school actions to reduce CALD disproportionality). Encourage specificity: "Not just 'translate resources' — WHERE would you distribute them? WHEN? WHO would deliver them?"

 EVALUATE

3 MIN

- Exit: "Name ONE social determinant that a school can address and ONE that it cannot. Why is that distinction important?"

 ENGAGE

6 MIN

- Open with: "The government has \$5 million to spend on drowning prevention. You have 60 seconds to decide how to spend it. Go." Quick-fire round — write answer on a piece of paper, then reveal and discuss.
- This primes Activity 4.1 (prevention strategy evaluation table and Q4.1a justification).

 EXPLORE

18 MIN

- Students complete Workbook Activity 4.1 (prevention strategy evaluation) — requires research and judgement, not just description.
- Students research for Activity 5.1 (international comparison table) using three organisation websites: royallifesaving.com.au, rlss.org.uk/education, redcross.org/take-a-class/water-safety.

 EXPLAIN

10 MIN

- Class share: What did students find about international approaches? Key comparison points: UK uses "Float to Live" campaign; Red Cross uses "Think So You Don't Sink" decision framework; RLSSA uses the "Water Safety Framework."
- Teacher highlights: what Australia does well (lifeguard system at patrolled beaches) vs areas for improvement (pool fencing compliance, CALD engagement).

 ELABORATE

12 MIN

- Students complete Q5.1a (identify one unadopted international element — argue whether Victoria should adopt it).
- Prompt: "Evidence-based argument means citing the organisation, not just saying 'I think this is good.'"

 EVALUATE

4 MIN

- Think-pair-share: "Which ONE prevention strategy has the strongest evidence, and which has the weakest? Defend your answer."

 ENGAGE

7 MIN

- Display the six-stat data grid (10%, 20%, 95%, 75%+, 66%, +34%) with no context. Ask: *"What do you think these numbers describe?"* Take 4–5 responses before revealing the topic.
- This cold-open approach consistently produces high engagement — students are surprised the topic is not drowning.
- Key question to pose: *"If 95% of patients are male — is that a biology fact or a social fact? What's the difference?"*

 EXPLORE

18 MIN

- Students read the three Australian case studies (Terrigal Wharf NSW 2025, Torquay Front Beach Vic 2025, Bawley Beach NSW 2024) individually — 5 minutes.
- Pair discussion (4 min): *"What single factor appears in all three cases? Why is that factor particularly dangerous?"* Lead students toward false confidence and depth variation — not just 'they dived into shallow water.'
- Class discussion (4 min): Draw out the cervical spine mechanism. If you have a foam spine model or skeleton, this is the ideal time to use it. The key physiological point: C4–C6 compression → instant permanent quadriplegia.
- Introduce inland waterway risk factors using the five-factor framework. Students add notes to their workbook.

 EXPLAIN

8 MIN

- Direct instruction (5 min): Walk through the social determinants explanation for the 95% male statistic. Use the masculinity norms framework from Section 3. The argument students need to be able to make: *this is not biology, it is a social construction that can be changed.*
- Introduce the concept of false confidence as a structural problem, not an individual attitude. Reference the Bawley Beach sandbank formation — the person did everything they had learned was safe. The information environment failed them.

 ELABORATE

12 MIN

- Students complete Activities 6.1 and 6.2 (case study analysis and myth vs evidence) in the workbook. Circulate and prompt deeper reasoning — push students away from surface-level "they should have been more careful" responses.
- Activity 6.3 (prevention strategy design) can begin in class or be set as homework preparation for Lesson 6.
- For extension students: prompt them to locate one primary source from AIHW or RLSSA before next lesson.

 EVALUATE

5 MIN

- Exit ticket: *"What is one specific way the water depth at Bawley Beach could have changed between the last safe visit and the injury — and what should have been done differently?"*
- Correct answer involves sandbank formation and the need to check depth physically every time, regardless of prior experience.

Sensitivity Note

If you have students who have experienced aquatic injury (their own or a family member's), be aware this content can be confronting. The case studies are factual but involve serious injury. Brief students at the start that the content involves real injuries, and have a plan for any student who needs to step out.

6

ENGAGE

8 MIN

- Rapid revision Kahoot covering Lessons 1–4 content (use Question Bank resource for Y9–10 equivalent questions). Focus on: rate calculations, physiological terms, social determinants, prevention strategy hierarchy.

ELABORATE

32 MIN

- **Option A — Policy Brief:** 600–800 words to Victorian Minister for Sport. Must cite ≥ 3 sources with URLs, use Victorian data, address a social determinant, and propose measurable outcomes. Scaffold: Situation → Evidence → Recommendations ($\times 2$) → Success Metrics.
- **Option B — Multimodal Campaign:** Target ONE high-risk group from Victorian data. Must include: campaign name, key message (evidence-based), visual concept description, target platform, and 100-word justification of design decisions.
- **Option C — School Audit:** Structured report on the school's current water safety provision. Must benchmark against Curriculum 2.0 requirements (VC2HP8P10, VC2HP8M01) and LSV classroom resources toolkit. Include a priority matrix (urgency $\times$ impact).

EVALUATE

10 MIN

- Peer review using the Assessment Checklist (Section 6 of this guide). Partners check: evidence present? Audience clear? Call to action specific? Sources cited with URLs?
- Collect completed workbooks. Return with teacher feedback before report writing is finalised.

✦ Answer Key Guidance

Year 9–10 questions are largely analytical and evaluative — there is no single correct answer for most questions. Use the sample answers as **quality benchmarks** for the A-level standard. Accept well-reasoned alternatives with supporting evidence.

Section 6 — Diving, Spinal Injuries & High-Risk Behaviour

Activity 6.1a (common factors): Strong answers identify *false confidence from prior experience* and *unknown/changing water depth* as the common mechanisms — not just "they dived." The mechanism matters: depth changed due to sandbank formation (Bawley), tidal variation (Torquay), and structural shallow zone beneath a wharf (Terrigal).

Activity 6.1b (95% male — structural explanation): A-standard responses use the social determinants framework from Section 3: masculinity norms, peer pressure in group settings, social reward for risk-taking, and suppression of fear responses. Reject responses that cite "males are naturally more reckless" — this is a biological reductionism error.

Activity 6.1c (cervical spine → quadriplegia): The key physiological chain: C4–C6 fracture or dislocation → spinal cord compression or severance → immediate loss of motor and sensory function below the injury level. Spinal cord neurons cannot regenerate. 66% of patients suffer permanent disability.

Activity 6.2 (myth refutation): Mark for mechanism, not just assertion. "That is a myth because depth can change" is C-standard. "That is a myth because sandbank formation, tidal variation, and sediment accumulation can alter depth between visits, as documented at Bawley Beach in 2024" is A-standard.

SECTION 1 — Q1.1A: MALE % CALCULATION

Sample answer (A standard): $46 \div 54 \times 100 = 85.2\%$. This is higher than the global figure of ~80%, suggesting Victorian males take proportionally more risks near water than the global average male population. This may reflect Victorian cultural norms around risk-taking, the prevalence of unpatrolled inland waterways accessible to males (fishing, kayaking, swimming), and the influence of peer pressure and alcohol in recreational settings.

SECTION 1 — Q1.1B: RATE COMPARISON CALCULATION

Calculation: $1.2 \text{ per } 100,000 \times 69 \text{ (units of } 100,000 \text{ in } 6.9\text{M)} = 82.8$ expected deaths at national average. Lives saved = $54 - 83 = -29$ (Victoria is actually BELOW the national average). **Key insight:** Victoria performs better than the national average overall, but has specific problem areas (CALD communities, inland waterways) that pull the figure up. A student who calculates and recognises this nuance is working at A standard.

SECTION 1 — Q1.1C: CALD OVER-REPRESENTATION ANALYSIS

Sample answer (A standard): CALD Victorians are significantly over-represented — they account for 40% of drowning deaths but only ~28% of the population. The over-representation ratio is approximately 1.43:1. Two reasons beyond "language barriers": (1) Many CALD community members grew up in countries without access to swim lessons or public pools — the skill gap is structural, not individual. (2) Higher participation in high-risk water activities (commercial fishing) with less safety training and equipment than recreational users.

SECTION 2 — Q2.1A: PHYSIOLOGICAL SEQUENCE OF DROWNING

Sample sequence (A standard): 1. Submersion → breath-holding/panic. 2. Laryngospasm (involuntary closure of airway — accounts for "dry drowning" ~10–15% of cases). 3. In wet drowning: laryngospasm fails, water enters lungs → hypoxia begins. 4. Hypoxia → loss of consciousness. 5. Cardiac arrest. 6. Cerebral hypoxia begins within 4–6 minutes — brain cells begin dying irreversibly. 7. Biological death. *Students must use all five listed terms to achieve full marks.*

SECTION 2 – Q2.1B: WHY EARLY CPR MATTERS

Sample answer (A standard): CPR generates approximately 25–30% of normal cardiac output. While insufficient for normal functioning, this is enough to deliver oxygenated blood to the brain and heart muscle at a rate that prevents irreversible cell death. After 4–6 minutes without oxygenated blood, neurons begin dying. After 10 minutes without CPR, the probability of neurologically intact survival drops dramatically. Starting CPR within 4 minutes — even by an untrained bystander — maintains minimum oxygenation until defibrillation or ROSC (return of spontaneous circulation).

SECTION 3 – Q3.1A: SOCIAL DETERMINANTS + PREVENTION STRATEGIES

Accept any two well-reasoned determinants from: gender norms (masculinity → risk-taking), CALD background (limited swim access, fishing risk, language barriers), socioeconomic status (less likely to have learned to swim as children; limited access to patrolled beaches), rural/regional location (higher proportion of unpatrolled inland waterway use), age (15–24 risk-taking peak). **Prevention strategy** must specifically target the mechanism — e.g., for masculinity norms: male-specific messaging that reframes water safety as strength (knowing when to walk away), not weakness.

SECTION 4 – Q4.1A: \$5M PREVENTION STRATEGY JUSTIFICATION

Sample answer (A standard): Priority 1 — Expand patrolled beach hours and locations (strongest evidence base: 0 deaths at patrolled sites 2023–24; direct intervention against the highest-risk behaviour). Priority 2 — Targeted CALD swimming access programs: subsidised swimming lessons delivered in-language by CALD community members (addresses structural access gap; has demonstrated impact in RLSSA pilot programs). *Accept any well-justified combination with evidence from the prevention strategy table.*

SECTION 5 – Q5.1A: INTERNATIONAL ELEMENT VICTORIA SHOULD ADOPT

Sample answer (A standard): The UK RLSS "Float to Live" campaign (rlss.org.uk) explicitly teaches floating as the PRIMARY response to any water emergency — simpler and more physiologically sound than attempting to swim. Victorian messaging focuses primarily on swimming to safety. Given that floating requires no fitness, works in most water conditions, and directly counteracts the panic-induced flailing that leads to exhaustion and drowning, adopting this as the core survival message would complement rather than replace existing SwimSafe rules.

5 Assessment Rubric — A–E, 6 Criteria

How to Use This Rubric

This rubric applies to both the **workbook** (collected end of unit) and the **summative task** (Section 6, one of three options). The six criteria reflect the Year 9–10 achievement standard. Holistic judgement across criteria — not mechanical addition of marks — produces the most valid grade.

Criterion	E — Emerging	D — Developing	C — Consolidating	B — Building	A — Extending
1. Data Analysis & Calculation VC2HP10P08	Cannot perform basic drowning rate calculations; misinterprets data	Performs calculations with errors; limited interpretation of results	Correctly calculates rates; provides reasonable interpretation of figures	Accurate calculations; interprets significance and compares across contexts; identifies trends	Sophisticated calculations with full working; nuanced interpretation; identifies implications and limitations of data
2. Physiological Knowledge VC2HP10P08 · VC2HP10M01	Cannot describe drowning physiology; vocabulary absent or incorrect	Names some physiological processes; limited understanding of mechanisms	Describes drowning sequence accurately; uses most key terms correctly	Explains mechanisms with precision; connects physiology to CPR rationale and survival outcomes	Sophisticated, accurate physiological analysis; uses all specified terms correctly; explains clinical implications (secondary drowning, 4-min rule)
3. Social Determinants Analysis VC2HP10P02	Identifies individuals as responsible; no systemic analysis	Acknowledges social factors exist; limited analysis of mechanisms	Identifies key social determinants; explains how they increase risk	Analyses multiple determinants with specific evidence; proposes targeted strategies	Sophisticated structural analysis; distinguishes individual vs systemic causes; proposes evidence-based, equitable interventions
4. Prevention Strategy Evaluation VC2HP10P08 · VC2HP10P09	Cannot distinguish stronger from weaker prevention evidence	Names prevention strategies; limited evaluation of effectiveness	Evaluates strategies with reference to evidence; identifies limitations	Comparative evaluation using evidence hierarchy; justifies resource allocation decisions	Critical evaluation against evidence; acknowledges trade-offs; sophisticated prioritisation with specific justification
5. International Comparison VC2HP10P09	No research evident; cannot identify international approaches	Names one international approach; superficial description	Compares at least two international approaches against Victorian practice	Identifies substantive differences and similarities; evaluates relevance to Victorian context	Critical, evidence-based comparison; argues specifically for/against adoption of international elements; cites primary sources
6. Communication	Incomplete; no clear audience,	Basic message; vague	Clear message; appropriate	Compelling product;	Sophisticated, persuasive,

Product VC2HP10P08	evidence, or call to action	audience; one fact with weak citation	audience; at least one data- backed claim with source URL	evidence- based claims; audience- appropriate language; specific measurable call to action	evidence-rich product; demonstrates health literacy; measurable outcomes; addresses barriers specific to target group; primary source citations
------------------------------	--------------------------------	---	--	---	---

6 Assessment Checklist — Summative Task

How to Use This Checklist

Use alongside the rubric. Tick each item once confirmed present. The checklist ensures minimum requirements are met before holistic rubric grading is applied. Share with students before they submit — it functions as both a teacher marking guide and a student self-check.

All Options — Minimum Requirements

- ☐ At least 3 credible sources cited with full URLs
- ☐ At least 1 accurate Victorian or global statistic with source
- ☐ Clear identification of target audience or addressee
- ☐ A specific, measurable call to action or recommendation
- ☐ Evidence of drafting/planning in workbook planning space
- ☐ Connection to at least one social determinant or physiological concept from unit

Option A — Policy Brief Checklist

- ☐ Addressed to Victorian Minister for Sport (or equivalent)
- ☐ 600–800 words (or equivalent scope)
- ☐ Two specific recommendations clearly stated
- ☐ At least one counter-argument acknowledged and addressed
- ☐ Measurable success metric for each recommendation
- ☐ Victorian data cited (LSV 2023–24 or equivalent)
- ☐ References Victorian Curriculum or LSV/RLSSA framework

Option B — Multimodal Campaign Checklist

Option C — School Audit Checklist

- ☐ ONE specific high-risk group identified from Victorian data
- ☐ Campaign name and key message clearly stated
- ☐ Visual concept described or shown
- ☐ Target platform specified (Instagram, TikTok, school notice boards, etc.)
- ☐ 100-word justification of design decisions
- ☐ Addresses a specific barrier for the target group
- ☐ Evidence cited for the barrier addressed

- ☐ Current provision described (what the school does now)
- ☐ Gaps identified with reference to Curriculum 2.0 codes
- ☐ Benchmarked against LSV Classroom Toolkit
- ☐ At least 3 specific recommendations with priority rationale
- ☐ Priority matrix included (urgency × impact, or similar)
- ☐ Realistic implementation suggestions (not just aspirational)
- ☐ 400-word minimum (or equivalent scope)

Report Comment Stems — Year 9–10 Water Safety

- **A standard:** "[Name] demonstrated sophisticated analytical skills in interpreting Victorian and global drowning data, producing a highly persuasive and evidence-rich [policy brief / campaign / audit] that clearly addressed the needs of a specific at-risk group."
- **C standard:** "[Name] showed sound knowledge of drowning prevention strategies and social determinants of health, producing a competent [product] that cited relevant data and identified a clear call to action."
- **E standard:** "[Name] is developing their understanding of evidence-based health analysis. With additional support, [Name] will build confidence in using data to justify health recommendations."

7 Differentiation & Inclusion

Extension Students

- Assign the Extension Workbook (Year 7–8 level, deeper topics) alongside this unit for cross-year comparison tasks
- Option C (School Audit) is the most complex assessment task — recommend for highly capable students
- Encourage use of WHO and CDC international data sets for Section 1 calculations
- Research task: compare Victorian drowning prevention funding as a % of health budget vs

Students Requiring Support

- Provide the calculation scaffold for Section 1 (rate = deaths ÷ population units × 100,000)
- Allow verbal responses for physiological sequence questions (Q2.1a, Q2.1b)
- For Section 3 social determinants: provide a pre-populated list of determinants — students choose two and explain, rather than generate their own list
- Option B (multimodal campaign) is the most accessible assessment task — recommend for

UK/USA

- Invite a guest speaker: LSV community engagement officer, RLSSA researcher, or local council water safety coordinator

students with lower literacy

- Pair with a peer research partner for the international comparison activity

CALD Students

- The CALD over-representation data (Section 1) may be personally significant — frame as a systemic issue, not an individual one
- Option C (School Audit) may surface valuable community knowledge — CALD students' experience is data too
- LSV provides translated materials that can be referenced in student assessment tasks: lsv.com.au/community/multicultural-safety/
- For Option B: encourage targeting their own community — this is the most impactful application of the unit

EAL/D Students

- Provide a vocabulary glossary: hypoxia, laryngospasm, cardiac output, determinants, oxygenation, cerebral
- Allow diagram/annotated visual as an alternative to written physiological sequence (Q2.1a)
- Assessment task: allow mixed language (home language + English) for planning notes; final product in English
- Sentence starters for the policy brief option: "The evidence suggests...", "Specifically, I recommend...", "This strategy would address..."

8 Teacher Background Notes

Physiology — Quick Reference for Teachers

- **Hypoxia:** Oxygen deprivation at tissue level. Cerebral hypoxia (brain) is the critical mechanism in drowning fatality. Begins within 4 minutes of cardiac arrest.
- **Laryngospasm:** Involuntary closure of the vocal cords on water contact. Causes "dry drowning" (~10–15% of drowning deaths). Water never enters lungs, but victim still asphyxiates.
- **CPR cardiac output:** Standard chest compressions at 100–120 bpm, 5–6cm depth, generate ~25–30% of normal resting cardiac output — enough to prevent brain death for 10–15 minutes pending defibrillation.
- **Cold water shock:** Below ~15°C triggers involuntary gasp (aspiration risk), hyperventilation, peripheral vasoconstriction (cold blood from extremities → heart), and possible cardiac arrhythmia. Can incapacitate within 1 minute.
- **Secondary drowning:** Pulmonary oedema (fluid in lungs) developing hours after near-drowning. Symptoms: coughing, fatigue, chest pain. Requires medical review. Rare but potentially fatal.

Diving & Spinal Injury — Teacher Content Background

The **cervical spine (C1–C7)** is the most vulnerable region in a diving injury because the full force of impact travels up through the arms and shoulders into the neck. At C4–C6, compression or dislocation can interrupt the spinal cord's signals to the diaphragm (C3–C5 innervates breathing muscles), the arms, and the legs simultaneously.

Key statistics to know: 10% of all spinal cord injuries in Australia result from unsafe diving; 20% of quadriplegia cases; 95% of patients male; 75%+ under 30; 66% permanent disability; highest incidence age group 10–14 (AIHW).

The **"Feet First, First Time" message** is Australia's core prevention response. Its limitation is that it addresses knowledge (what to do) but not the social norms (why teenagers don't) — which is why the structural explanation is pedagogically important at Year 9–10 level.

Sources for your own reading: aihw.gov.au (search "aquatic injury") · royallifesaving.com.au (spinal injury) · Royal Perth Hospital spinal injury study (academic databases)

✦ Curriculum 2.0 Implementation — Year 9–10

At Year 9–10, the critical move is from **understanding** (can explain what and why) to **evaluation and synthesis** (can judge quality of evidence and produce new health products). This unit is specifically designed to scaffold that shift: Sections 1–3 build analytical vocabulary; Sections 4–5 require judgement; Section 6 demands synthesis.

- Reference: victoriancurriculum.vcaa.vic.edu.au/health-and-physical-education/curriculum/f-10
- DE Swimming Policy: www2.education.vic.gov.au/pal/swimming-instruction-and-water-safety-education/policy

9 Resources & Sources

Life Saving Victoria — Drowning Reports

<https://lsv.com.au/research/victorian-drowning-reports/> · Primary data source for all Victorian statistics

LSV Classroom Toolkit

https://lsv.com.au/toolkit/classroom_based_water_safety.php · Free lessons aligned to Victorian Curriculum 2.0

LSV Multicultural Safety Resources

<https://lsv.com.au/community/multicultural-safety/> · Translated materials for CALD engagement

Royal Life Saving Society Australia

<https://www.royallifesaving.com.au> · National drowning data, evidence-based prevention frameworks

World Health Organisation — Drowning Fact Sheet

<https://www.who.int/news-room/fact-sheets/detail/drowning> · Global epidemiological data for Section 1 comparisons

RLSS UK — Education Resources

<https://www.rlss.org.uk/education> · "Float to Live" campaign; international classroom comparison

American Red Cross — Water Safety

<https://www.redcross.org/take-a-class/water-safety> · "Think So You Don't Sink" framework; international comparison

Victorian Curriculum (VCAA)

<https://f10.vcaa.vic.edu.au> · Official content descriptions and achievement standards

Paddle Australia

<https://paddle.org.au> · Inland waterway safety; river grading; Victorian river context

CDC Drowning Prevention

<https://www.cdc.gov/drowning/prevention> · USA data for international comparison tasks

Advanced Water Safety — Teacher Resource Guide · Victorian Curriculum F–10 Version 2.0 · HPE Year 9–10 · VC2HP10P02, VC2HP10P08, VC2HP10P09, VC2HP10M01

lsv.com.au · royallifesaving.com.au · who.int · victoriancurriculum.vcaa.vic.edu.au